most programming languages, and usually these libraries are written by experts that knows how to optimize performance or take care of security concerns.

The main goal of this book, in fact, is to give you familiarity with a broad range of tools, and train you to recognize opportunities to use them to improve your code. Understanding how this tools work internally, at least at a high level, is an important part of the learning process. But nevertheless, there are still certain situations in which you might need to get your hands dirty with code, for example if you are using a brand new programming language for which there aren't many libraries available, or if you need to customize a data structure to solve a special case.

In the end, whether you are going to write your own implementation of data structures really depends on many factors.

First and foremost, how advanced is the data structure you need, and how mainstream is the language you use.

To illustrate this point, let's take clustering as an example.

If you are working with a mainstream language, for instance in Java or Python it's very likely that you can find many trusted libraries for k-means, one of the simplest clustering algorithms.

If you are using a niche language, maybe experimenting with a recently created one like *Nim* or *Rust*, then it might be harder to find an open source library implemented by a